

Q 30.

- I2C (Inter-Integrated Circuit) is a communication protocol used to connect multiple devices (like sensors) to a microcontroller using only two wires.
It is widely used in IoT systems for efficient and reliable data communication.
- SDA and SCL lines :
 - SDA (Serial Data Line): Used to transfer data between the master and slave devices.
 - SCL (Serial Clock Line): Used to synchronize the data transfer between devices.
- I2C addressing :
 - Each I2C device has a unique 7-bit address (0x03 to 0x77).
 - The master device (like Arduino) sends the address of the slave device it wants to communicate with.
 - The slave with that address responds, and communication begins.
- Three common IoT sensors that use I2C :
 - MPU6050 (Accelerometer + Gyroscope)
 - BME280 (Temperature, Humidity, Pressure)
 - OLED Display (SSD1306)